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## Team 23435 Gyrobotic Droids Shooter Calcs Explained

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One of our goals this season was to shoot reliably, anywhere on the field, and while the robot was in motion. Our robot design is a rotating turret on a mecanum drive base with an adjustable angle shooter hood. At a high level, we approached this problem as follows:

- A. Calculate Initial Ball Launch Angle and Velocity: We want the ball to enter the goal at a certain position at a certain angle.
- B. Velocity Compensation: Compensate the turret position, launch angle, and launch velocity based on the robot's current velocity on the field
- C. Convert Launch Velocity to Flywheel Speed and Launch Angle to Hood Servo Position

### A. Calculate Initial Ball Launch Angle and Velocity

Generated with the help of AI.

#### Projectile Motion with a Required Angle at a Point

We want to find **how fast** and **at what angle** to launch a ball so that:

- It is launched from the origin (0,0) – defined as our robot turret exit
- It passes through a point (x,y) – which is the position of the goal
- At that point, it is moving at a specific angle  $\theta$  (relative to the horizontal)

We assume:

- Gravity is constant:  $g = 9.8 \text{ m/s}^2 = 386.1 \text{ in/s}^2$
- No air resistance
- Robot is stationary (for now, velocity compensation will follow)

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#### 1. Break the motion into horizontal and vertical parts

Projectile motion is easiest when split into **horizontal** and **vertical** motion.

Launch quantities (both are unknown at this point)

- Initial speed:  $v_0$
- Launch angle:  $\alpha$

Velocity components

$$v_x = v_0 \cos \alpha$$

$$v_y = v_0 \sin \alpha - gt$$

Position components (integrate the velocity equations)

$$x = v_0 \cos \alpha t$$

$$y(t) = v_0 \sin \alpha t - \frac{1}{2}gt^2$$

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#### 2. Trajectory angle at any point

The trajectory angle at any point is the change in y divided by the change in x, which simplifies to:

$$\tan \theta = \frac{v_y}{v_x}$$

Plugging the velocity components in the above formula:

$$\tan \theta = \frac{v_0 \sin \alpha - gt}{v_0 \cos \alpha}$$

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### 3. Eliminate time using the horizontal position formula for trajectory angle

From the horizontal position, x, solve for t:

$$x = v_0 \cos \alpha t \Rightarrow t = \frac{x}{v_0 \cos \alpha}$$

Substitute into the angle equation:

$$\tan \theta = \frac{v_0 \sin \alpha - g \left( \frac{x}{v_0 \cos \alpha} \right)}{v_0 \cos \alpha}$$

Split the fraction:

$$\tan \theta = \frac{v_0 \sin \alpha}{v_0 \cos \alpha} - \frac{gx}{v_0^2 \cos^2 \alpha}$$

Recognize the first term: ( $v_0 \sin(\alpha)$  = vertical velocity at launch,  $v_0 \cos(\alpha)$  = horizontal velocity at launch) vertical launch velocity / horizontal launch velocity =  $\tan(\text{launch angle})$

$$\frac{v_0 \sin \alpha}{v_0 \cos \alpha} = \tan \alpha$$

So we get:

$$\tan \theta = \tan \alpha - \frac{gx}{v_0^2 \cos^2 \alpha}$$

Rearrange the formula for  $v_0^2$ : (useful for later)

$$v_0^2 = \frac{gx}{\cos^2 \alpha (\tan \alpha - \tan \theta)}$$

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### 4. Eliminate time using the horizontal position formula for y position

From the horizontal position, x, solve for t:

$$x = v_0 \cos \alpha t \Rightarrow t = \frac{x}{v_0 \cos \alpha}$$

Substitute into the y position equation:

$$y(t) = v_0 \sin \alpha t - \frac{1}{2}gt^2$$

$$y = v_0 \sin \alpha \left( \frac{x}{v_0 \cos \alpha} \right) - \frac{1}{2}g \left( \frac{x}{v_0 \cos \alpha} \right)^2$$

$$= x \tan \alpha - \frac{gx^2}{2v_0^2 \cos^2 \alpha}$$

### 5. Combine the final formulas from step 3 and 4 above to get launch angle

Substitute  $v_0^2$  from step 3 into the final step 4 equation for vertical position:

$$y = x \tan \alpha - \frac{gx^2}{2v_0^2 \cos^2 \alpha}$$

Working with just the second term that has the  $v_0^2$ :

$$\frac{gx^2}{2v_0^2 \cos^2 \alpha} \rightarrow \frac{gx^2}{2 \left( \frac{gx}{\cos^2 \alpha (\tan \alpha - \tan \theta)} \right) \cos^2 \alpha}$$

Rewrite the denominator carefully:

$$2 \cdot \frac{gx}{\cos^2 \alpha (\tan \alpha - \tan \theta)} \cdot \cos^2 \alpha$$

Now notice something important:

$$\cos^2 \alpha \text{ appears in the numerator and denominator}$$

So they cancel out, leaving:

$$2gx / (\tan \alpha - \tan \theta)$$

Now the full expression becomes:

$$\frac{gx^2}{\frac{2gx}{\tan \alpha - \tan \theta}} = gx^2 \cdot \frac{\tan \alpha - \tan \theta}{2gx} = \frac{x}{2}(\tan \alpha - \tan \theta)$$

Substitute back into the full position equation:

$$y = x \tan \alpha - \frac{x}{2}(\tan \alpha - \tan \theta) = x \tan \alpha - \frac{x}{2} \tan \alpha + \frac{x}{2} \tan \theta$$

$$y = \frac{x}{2} \tan \alpha + \frac{x}{2} \tan \theta = \frac{x}{2}(\tan \alpha + \tan \theta)$$

Multiply both sides by  $2/x$ :

$$\frac{2y}{x} = \tan \alpha + \tan \theta$$

Subtract  $\tan(\theta)$ :

$$\tan \alpha = \frac{2y}{x} - \tan \theta$$

So the launch angle is:

$$\alpha = \arctan\left(\frac{2y}{x} - \tan \theta\right)$$

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## 6. Solve for the initial speed

Once  $\alpha$  is known, we use the position equation again:

$$y = x \tan \alpha - \frac{gx^2}{2v_0^2 \cos^2 \alpha}$$

Solve for  $v_0$ :

$$v_0 = \sqrt{\frac{gx^2}{2 \cos^2 \alpha (x \tan \alpha - y)}}$$

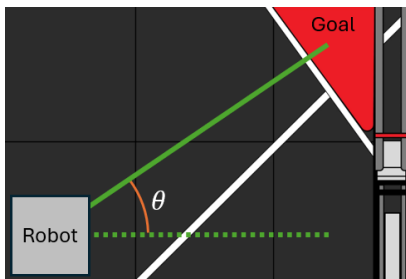
## B. Velocity Compensation

With the robot in motion, we need to do some vector math so that the ball will actually make it into the goal.

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### 1. Calculate the robot's velocity radially and tangentially to the goal.

Using the Pedro Pathing library, we get the robot's movement vector (which gives us speed and angle, so yes, you will want to use your odometry during teleop). We also need the angle of the vector from the robot to the goal.



First, calculate the difference between the above theta and the theta of the robot's velocity:

$$\theta = \theta_{velocity} - \theta_{line}$$

Then we can calculate, using the magnitude of the robot's velocity, the radial component:

$$- \cos(\theta) * V_{magnitude}$$

And the tangential component:

$$\sin(\theta) * V_{magnitude}$$

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## 2. Calculate the time it takes for the ball to get to the goal.

To do this, we use the horizontal velocity equation and solve for time:

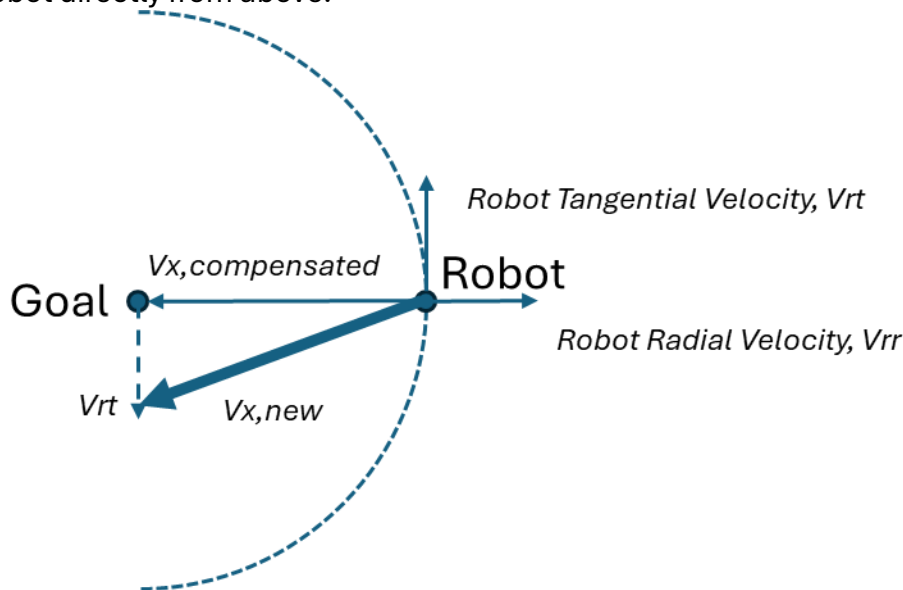
$$x = v_0 \cos \alpha t \Rightarrow t = \frac{x}{v_0 \cos \alpha}$$

Where x = distance to goal  
v0 and α from before

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## 3. Calculate a new ball speed in the x direction based on robot velocity.

Using the robot's tangential and radial velocity, calculate a new velocity the ball must travel in the x direction to get there in the same amount of time. To help with this, we view the robot directly from above:



First, compensate for the robot velocity in the radial direction simply by adding the ball x velocity and robot radial velocity together:

$$V_{x,compensated} = \frac{\text{Distance to Goal}}{\text{time}} + V_{rr}$$

And the compensation for the robot velocity in the tangential direction is simply Vrt.

So the hypotenuse is the Vx,new

$$V_{x,new} = \sqrt{V_{x,compensated}^2 + V_{rt}^2}$$

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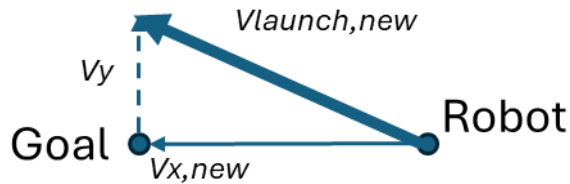
## 4. Calculate the launch velocity in the vertical, y, direction.

$$V_y = V_0 \sin(\alpha)$$

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## 5. Calculate the new ball launch angle based on the x and y components above.

To help visualize, view the robot from the side:



Using the arctan:

$$V_{launch,new} = \arctan\left(\frac{V_y}{V_{x,new}}\right)$$

At this stage, it may be important to clamp the output of this formula to something achievable for your robot. For example, it may want an angle of 35 degrees, but your robot can only go as low as 40 degrees.

## 6. Calculate the new ball launch speed based on the x and y components above.

First, calculate the new distance the ball travels in the x direction from  $V_{x,new}$  above:

$$x = V_{x,new} t$$

Then use the equation from above for launch speed:

$$v_0 = \sqrt{\frac{gx^2}{2 \cos^2 \alpha (x \tan \alpha - y)}}$$

Where g=gravity

x = the new distance

y = height of goal above robot launch point

$\alpha$  = launch angle from previous step (note, this may be the clamped angle)

## 7. Offset your turret angle for shooting

Viewing from above again, it's the angle of the new magnitude -  $V_{x,new}$ :

$$Turret\ Offset = \tan\left(\frac{V_{rt}}{V_{x,compensated}}\right)$$

## C. Convert Launch Velocity to Flywheel Speed and Launch Angle to Hood Servo Position

Now that we know the velocity compensated launch angle and launch speed, what speed do we command the flywheel and what position do we set the turret hood servo?

### 1. Hood Angle

We simply took 2 data points and calculated a line between the 2 points:

Min angle,  $a_1$  = Servo position,  $s_1$

Max angle,  $a_2$  = Servo position,  $s_2$

$$Servo\ Position = \left( \frac{s1 - s2}{a1 - a2} \right) (\alpha - a1) + s1$$

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## 2. Flywheel Speed

We took measurements where we varied the flywheel speed and hood angle (we manually set both of those) and measured how far the ball travels. Then we used the projectile range equation to calculate what the true launch speed was.

$$R = \frac{v_0^2 \sin(2\theta)}{g} \rightarrow v_0 = \sqrt{\frac{R \cdot g}{\sin(2\theta)}}$$

Then at various hood angles and various flywheel speeds, we used Excel to fit a line to get an equation that gives us flywheel speed as a function of launch speed